

Acknowledgement:

Part of the content was inspired and adapted from UC Berkeley CS61C Great Ideas of Computer Architecture (Machine Structures) Lab 5 Logisim Evolution.

Version info

Manual Version	Software Version	Author	Memo	Month/Year
V1.1	V3.8.0	Antian Wang	Update for ADA	Jun. 2026.
V1.0	V3.8.0	Antian Wang	Initial setup	Dec. 2023.

Software download:

Logisim Evolution can be downloaded from <https://github.com/logisim-evolution/logisim-evolution>

Go to the release link (<https://github.com/logisim-evolution/logisim-evolution/releases>) if you do not want to build the software by yourself.

According to the information at <https://github.com/logisim-evolution/logisim-evolution#download>, please download the appropriate installation package.

logisim-evolution-<version>-1_amd64.deb: Debian package (also suitable for Ubuntu and derivatives),

logisim-evolution-<version>-1.x86_64.rpm: Package for Fedora/Redhat/CentOS/SuSE Linux distributions,

logisim-evolution-<version>-amd64.snap: The Snap archive for all supported Linux distributions (also available in Snapcraft store),

logisim-evolution-<version>-aarch64.msi: Installer package for Microsoft Windows for Arm processors,

logisim-evolution-<version>-amd64.msi: Installer package for Microsoft Windows for Intel processors,

logisim-evolution-<version>-aarch64.dmg: macOS package for Apple processors,

logisim-evolution-<version>-x86_64.dmg: macOS package for Intel processors (also runs on Apple processors in simulation).

In most of the cases:

Windows user, please use logisim-evolution-<version>.msi

Mac user, please use logisim-evolution-<version>.dmg

Linux user, please use logisim-evolution-<version>-1.x86_64.rpm.

Please note that Logisim Evolution is open-source educational software. Some of your computers may prevent you from installing it.

For Windows users:

After seeing the warning sign, please click **MORE INFO** to see the install button.

For Mac users:

You need to open it from the download path for the first use. Since it is an open-source software, Apple may not allow you to open it by default. Follow the steps below to open the software. Click open if a pop-up window appears.

Logisim Evolution Tutorial

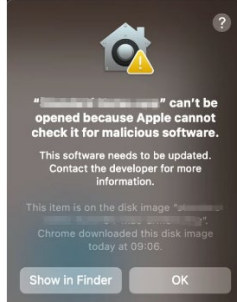


Figure 1 MacOS warning.

Apple can't check app for malicious software

Using an app that can't be checked for malicious software might harm your Mac or compromise your privacy. If you're certain that the app you want to use is from a trustworthy source, you can override your Mac security settings to open it.

1. In the Finder  on your Mac, locate the app you want to open.
Don't use Launchpad to locate the app.
 2. Control-click the app icon, then choose Open from the shortcut menu.
 3. Click Open.
- The app is saved as an exception to your security settings, and you can open it at any time by double-clicking it, just as you do with any registered app.

Figure 2 MacOS opens unknown software.

<https://support.apple.com/guide/mac-help/apple-cant-check-app-for-malicious-software-mchleab3a043/mac>

For Linux users:

Logisim Evolution requires Java 16 or newer.

Download the RPM file.

Get sudo privilege.

```
sudo alien -i xxxx.rpm
```

Known issue for Windows

Note that there may be an issue with Logisim Evolution, causing some download files to have a .circ extension. You can fix it using the method described at the link below.

<https://github.com/logisim-evolution/logisim-evolution/issues/1471#issuecomment-1332917265>

"Removing the registry key HKEY_CLASSES_ROOT\MIME\Database\Content Type\application/octet-stream\Extension instantly resolves the issue"

Known issue for Mac

TBD

Known issue for Linux

TBD

A tutorial accompanied by the software

In Menu->Help->tutorial, there is a complete list of features for Logisim Evolution. This tutorial covers only the features useful for this lab.

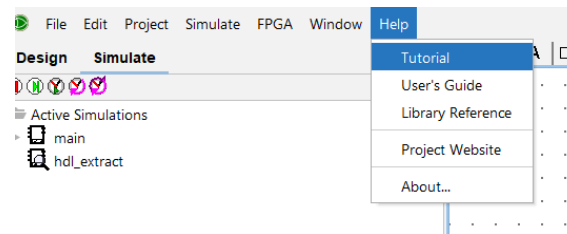


Figure 3 Access software tutorial.

Background

Logisim Evolution is the newer version of Logisim, which stopped its development in 2014. The Logisim Evolution has a better GUI and more advanced functionality than Logisim. It provides a strong logic simulation environment and helps you better understand the material in the undergraduate digital design course. This tutorial covers all Logisim Evolution features needed for the course. Please note that some Logisim Evolution features are beyond the scope of the undergraduate digital design course and may be more relevant to other ECE courses.

You can explore these features on your own for your own interest, but we only use the software's gate-level digital logic feature in this lab, including the final project.

Logisim Evolution usage

GUI interface

The GUI interface of the Logisim Evolution should look like the following. You can place as many logic gates as you want within the schematic on the right.

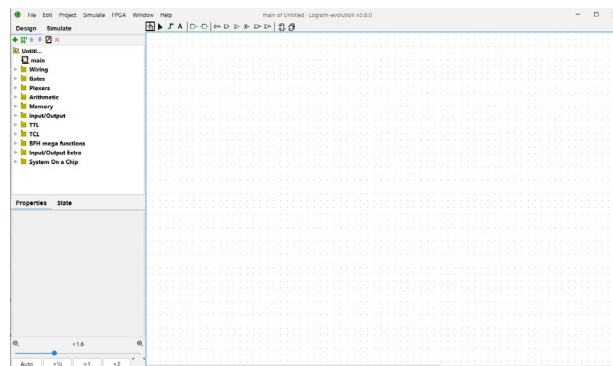


Figure 4 Logisim Evolution GUI interface.

GUI components

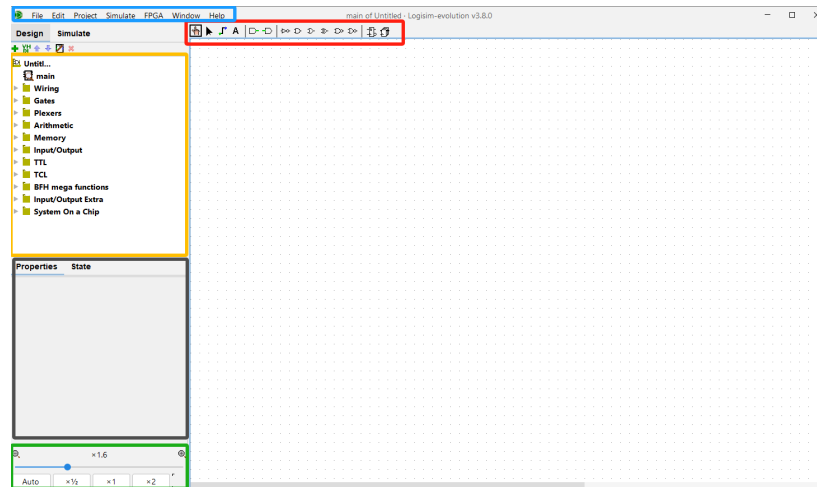


Figure 5 Logisim Evolution GUI region.

Basic usage shortcut

The red section provides shortcuts to typical editing tools, input/output pins, and simple logic gates.

When you approach the shortcut, there will be a brief hint & shortcut key beneath showing its meaning in case you are not familiar with them.

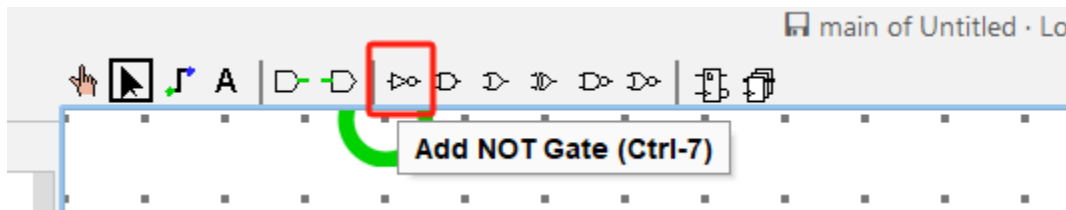


Figure 6 Shortcut toolbar.

Component library

The orange part shows the components that you can place in the schematic on the right.

You can expand the folder by clicking the triangle before it and finding the needed components.

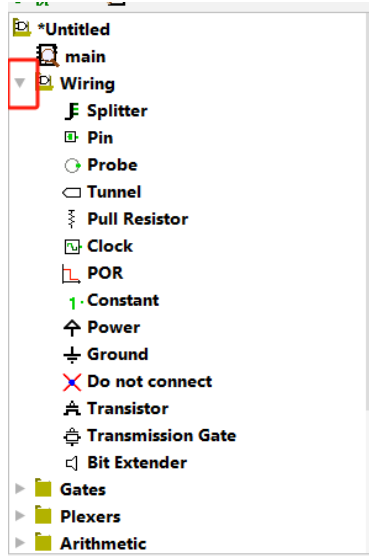


Figure 7 Expand the component folder.

Property & state editor

The black part will show more detailed properties of the components you place, which you could possibly edit.

For logic gates, you can use the property of negating one or multiple inputs to their complement. It avoids adding one additional inverter, and eases the wiring difficulty. Below is one simple example.

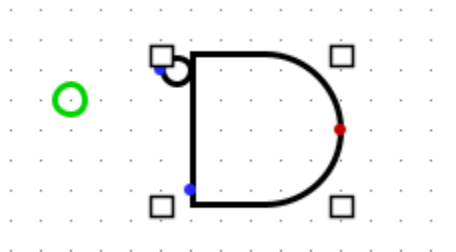


Figure 8 AND gate input negate.

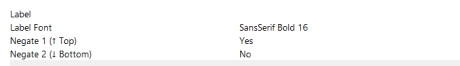


Figure 9 Invert the negate property setting.

Zoom in and zoom out

The green part allows you to zoom in and zoom out on the circuit diagram. You can also scroll up or down using your mouse by pressing Ctrl for Windows users, cmd for Mac users, and ctrl for Linux users to zoom in and zoom out.

If you happen to zoom in too much, you can re-center the design by clicking the “target” shown below.

Logisim Evolution Tutorial

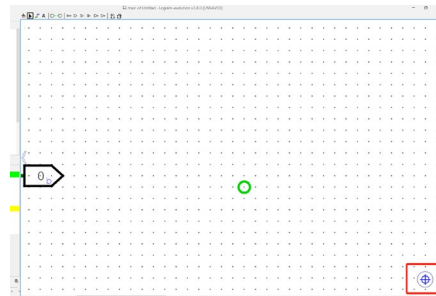


Figure 10 Recenter the schematic

And you can see the complete design within the schematic.

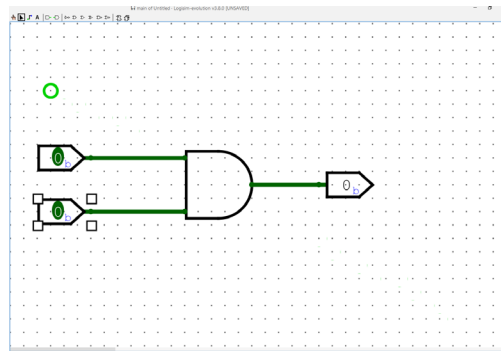


Figure 11 Schematic after recenter.

Basic project-wise setting

The blue part is like other GUI-based software that allows you to save and edit the project.

Toggling Input and Output

Input & output pins

The most used input/output (I/O) in the lab are input and output pins. There are two ways to find the two types: from the top of the shortcut toolbar, as shown below.

You are required to use input pins for your combinational input.

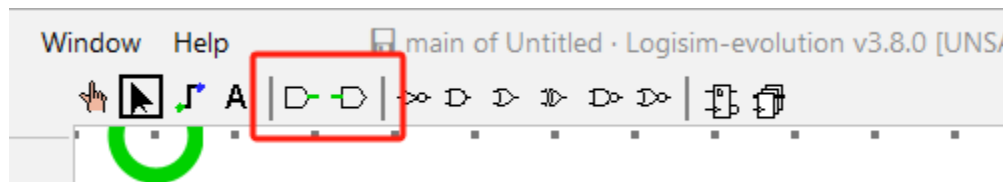


Figure 12 IO pin in the shortcut toolbar.

Or find it within the wiring section, and configure it to the needed function.

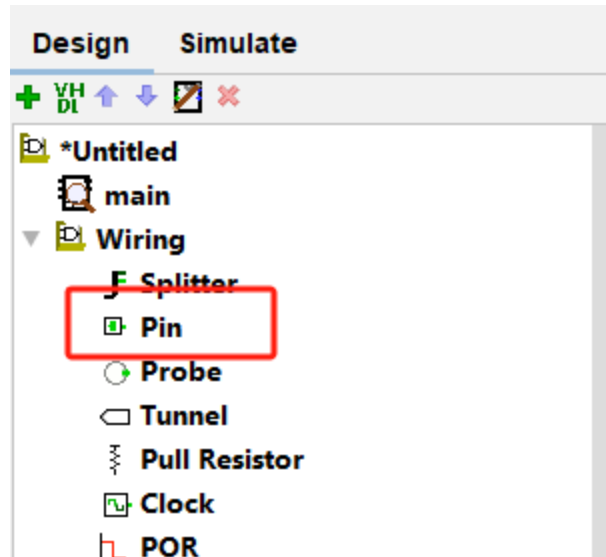


Figure 13 Pin in wiring folder.

Once you place the pin on the canvas, it shows the following properties to configure.

Properties	State
Pin (80,230)	
FPGA supported:	Supported
Facing	→ East
Output?	No
Data Bits	1
Three-state?	No
Pull Behavior	Unchanged
Label	HDL Required
Label Font	SansSerif Bold 16
Radix	Binary
Reset value:	0x0
Appearance	Arrow shapes

Figure 14 Pin's property.

Facing

You can change the facing of the components using your arrow keys on your keyboard or configure it here within the property. Below are the different views of the input pins, allowing you to make the design compact and organized.

Logisim Evolution Tutorial

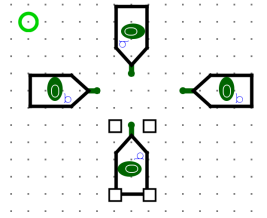


Figure 15 IO pins with different facing.

Output configuration

You can directly select the pins as output from the shortcut bar, but you can also configure it from input to output using the drop-down menu.

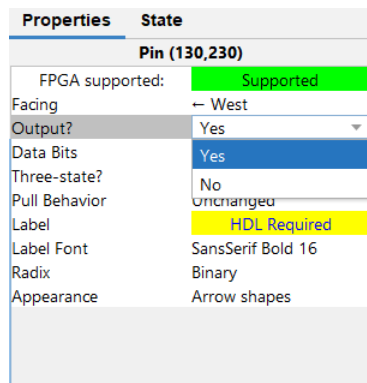


Figure 16 Configure the input pin to the output pin.

Data bit width selection

You can also configure the data bits' width by simply changing the data bits property. Below is what we look for a 5-bit input. Please be cautious about the bit-order.

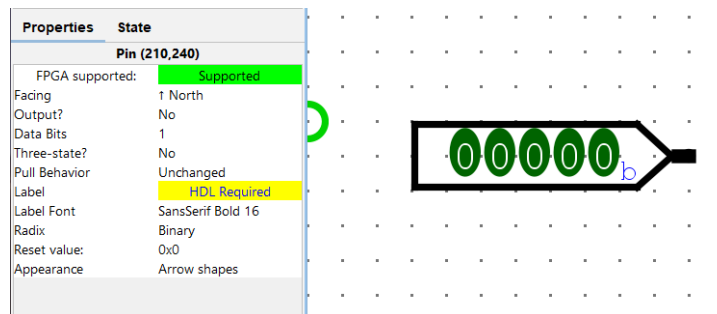


Figure 17 Configure to multi-bit pin.

Radix

Using a different radix may help you identify the value for multi-bit data, especially helpful for arithmetic logic designs.

Below is what it looks like by configuring the input pin to an unsigned decimal and connecting it to an output pin represented in binary. You can change the value by clicking the input pins.

Logisim Evolution Tutorial

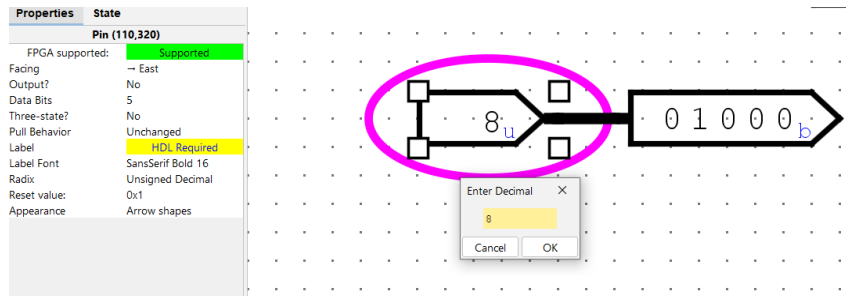


Figure 18 Using a non-binary radix pin.

DIP Switches

In most designs, you can use the DIP switch circuit input, located in the input/output folder, for convenient logic inputs. Below is the 8-bit DIP switch; you can configure it to whatever bit width you want.

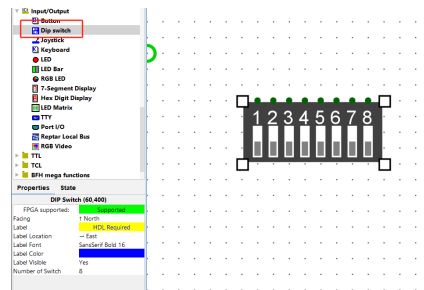
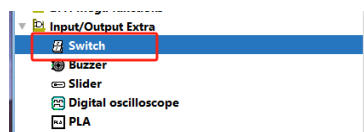


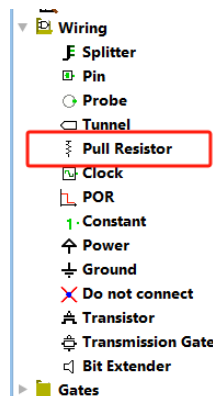
Figure 19 DIP switch.

Switch

In the input-output extra, there exists a “switch”, shown as below, different from the DIP switch above. However, switch is most similar to what you would wire in an actual circuit.



In such a setting, you need to use a new component named “pull resistor”.



If you configure the “pull resistor” to be pulled high, as shown below, you need to make the following connection.

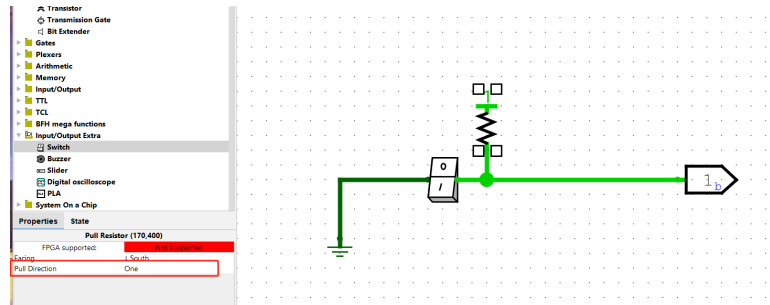


Figure 20 Pull up the resistor connection

In such a way, you can produce logic 0 & logic 1 by clicking the switch. If you configure the pull resistor to logic 0, then you need to use the following connection:

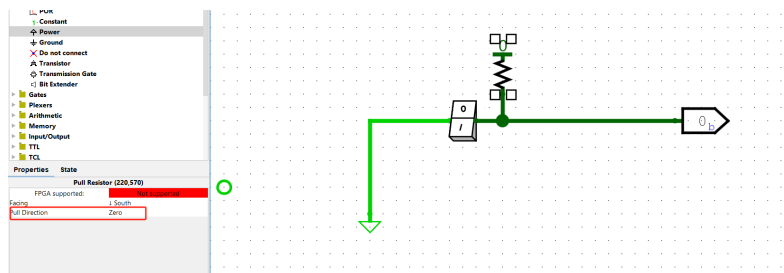


Figure 21 Pull down the resistor connection

LEDs

LED is another simple way to evaluate the output logic value. Here in Logisim Evolution, you can use the LED or an RGB LED based on your application choice.

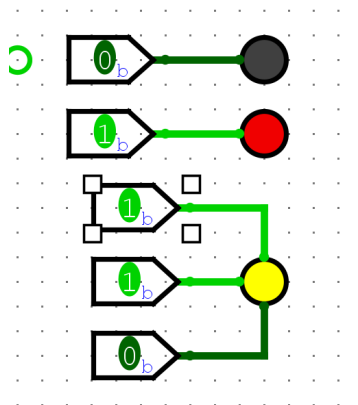


Figure 22 2-color LED and RGB LED.

Other input-output components

There are other interesting in-out components available to make your design more interactive, such as the LED bar and the LED matrix. You can explore it by yourselves.

Logic gate connection and configuration

Find the logic gate

You can find the logic gate at the top of the shortcut toolbar. You may also find additional logic gates under the gate folder.

Configure the logic gate input number

You can change the input numbers by changing the number of inputs, as shown by the 4-input AND gate on the right.

Configure the size of the logic gate

You can change the gate size to a narrower size to save space in the schematics shown on the left.

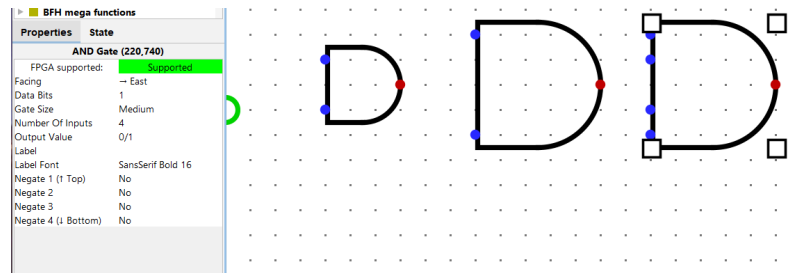


Figure 23 And a gate of a different size and input pins.

Verify the functionality

- Place the input/output/And gate on the schematic: Input  Output  And gate 

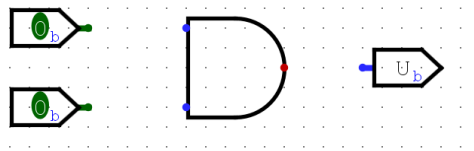


Figure 24 Connect and gates with two input pins and an output pin.

- Connect the pins together by 
- Use the  to toggle the pin's value.

You can then verify the functionality of the 2-INPUT AND gate.

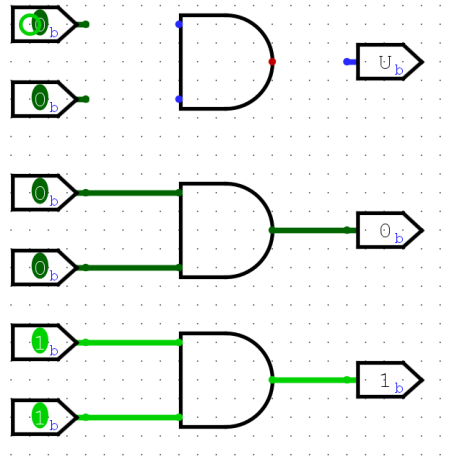


Figure 25 Step of the connection.

TTL IC configuration

One of the goals for the lab is to become familiar with 74-series logic chips. In Logisim Evolution, these gates are available under the TTL folder, with their IC number and corresponding functionality. It differs significantly from Logisim, where only custom 74 libraries are available and must be loaded each time.

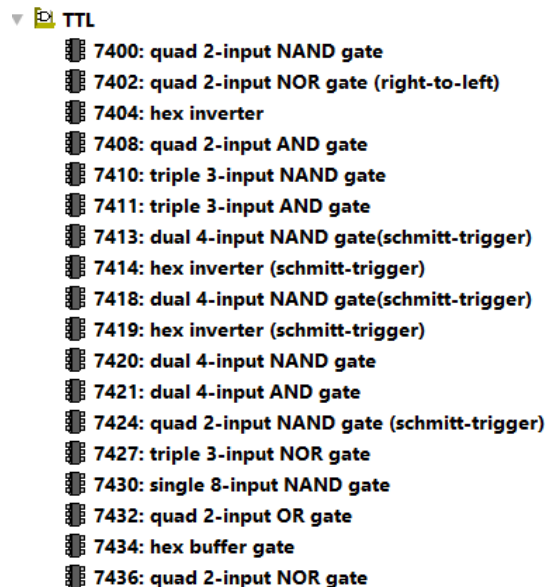


Figure 26 TTL 74 series IC folder

Find the IC in the menu

Here I use one of the 74 series TTL gates available, 7400, as an example.

Place IC on the schematic

Drag the gate to the schematic, and you will see the following:

Logisim Evolution Tutorial

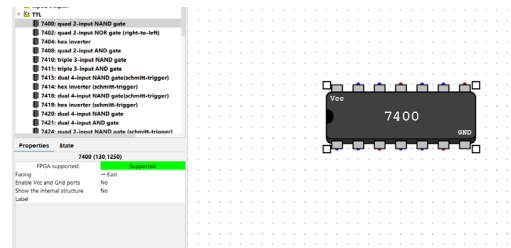


Figure 27 Place IC on schematic.

Configure the IC

There are two configurations that we typically encounter with most 74 series gates. Enable Vcc and Gnd ports and show the internal structure. My preference is not to enable the Vcc and Gnd ports, and to show the internal structure for clarity and simplicity.

Below is what it looks like when the internal structure is shown. The internal structure allows you to connect the IC without consulting the datasheet for the pin layout, making design and connection faster.

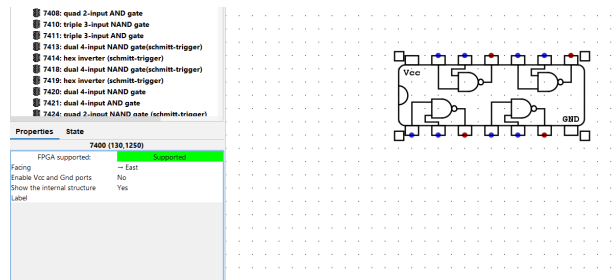


Figure 28 IC configuration with internal structure visible.

Wiring features

The following content is adapted from CS61C Lab 5.

Splitter

Splitters allow you to take a multi-bit value and split it up into smaller parts, or (despite the name) combine multiple values that are one or more bits into a single value. You can find it under the wiring folder. Please note that the splitter is bidirectional.

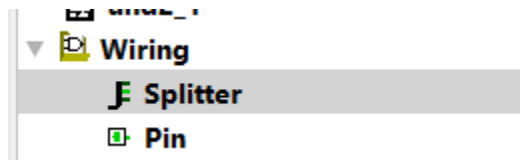
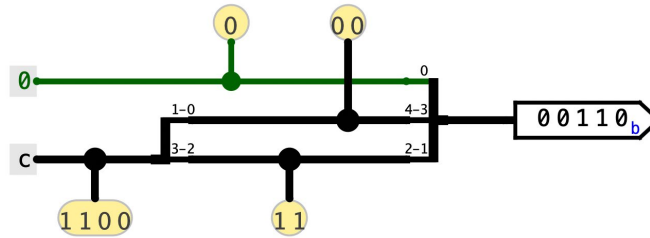


Figure 29 Location of splitter.

Here, we split the 4-bit binary number **1100** into **11** and **00**, swap their positions, and combine them with **0** to create the final 5-bit number **00110**:

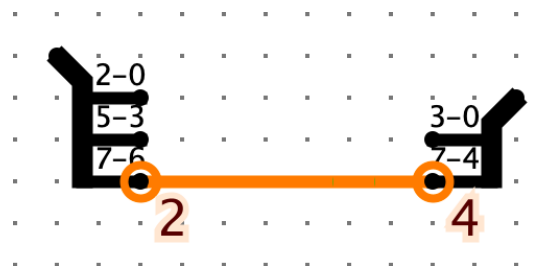


Click on a splitter (using the **Select** tool) to get its attributes in the sidebar (lower left). You can configure attributes such as the number of arms on your splitter and the number of bits on each arm. For the circuit above, the left and right splitters' attributes look like this:

Properties	State	Properties	State
Selection: Splitter		Selection: Splitter	
VHDL	Verilog	VHDL	Verilog
Facing	East	Facing	West
Fan Out	2	Fan Out	3
Bit Width In	4	Bit Width In	5
Appearance	Centered	Appearance	Centered
Spacing	2	Spacing	2
Bit 0	0 (Top)	Bit 0	0 (Top)
Bit 1	0 (Top)	Bit 1	2 (Bottom)
Bit 2	1 (Bottom)	Bit 2	2 (Bottom)
Bit 3	1 (Bottom)	Bit 3	1
		Bit 4	1

Notice that there's an attribute called **Facing**. You can use this to rotate your splitter. In the above example, the splitter on the right is facing **West** while the splitter on the left faces the **East**.

If you see an orange error wire, this means that your bit width in does not match your bit width out. Make sure that if you're connecting two components with a wire, you correctly set the bit width in that component's menu.



Tunnel

A tunnel allows you to draw an "invisible wire" to bind two points together. Tunnels are grouped by case-sensitive labels given to a wire. They are used to connect wires like so:

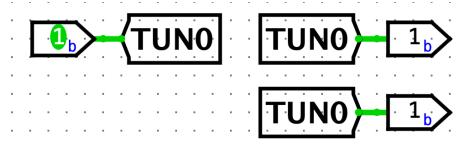


Figure 30 Tunnel example

which effectively is:



Figure 31 Equivalent tunnel connection

Here shows how you change the label of the tunnel.

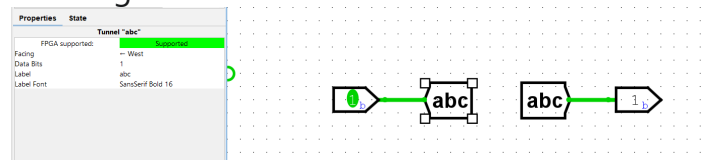
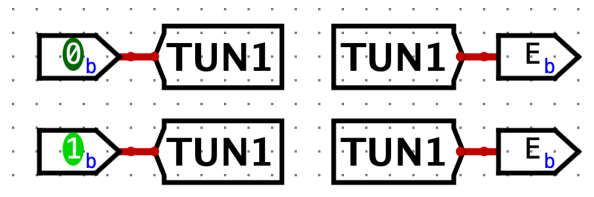
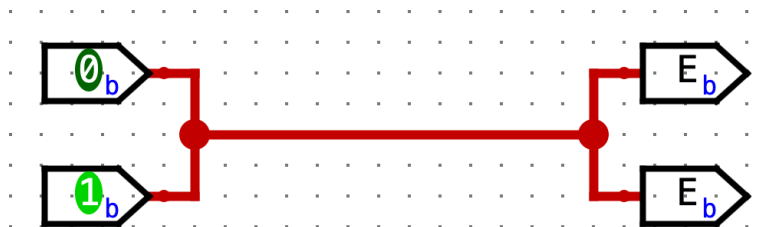


Figure 32 Change the label of tunnel.

Some care should be taken as to which wires are connected with tunnels to which other wires, such as in this case:



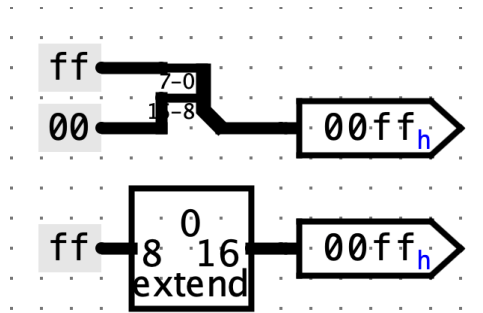
which effectively is:



We *strongly* recommend using tunnels in Logisim because they make your circuits look much cleaner (less wiring spaghetti), which makes debugging easier.

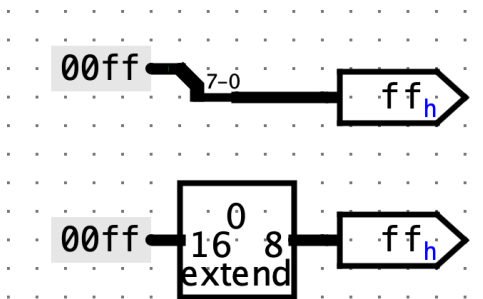
Extenders

When changing the width of a wire, you should use a bit extender for clarity. For example, consider the following implementation of zero-extending an 8-bit wire into a 16-bit wire:



Compared to the splitter, the extender is easier to understand at a glance. This becomes especially helpful when working with complex circuits.

Additionally, consider the case of throwing out bits. Despite its name, an extender can also perform this operation:



While the splitter is more minimalistic, it's also slightly harder to read.

Wire connection warning

Logisim Evolution provides a highly interactive color representation of connections. Please refer to the graph and table below for details.

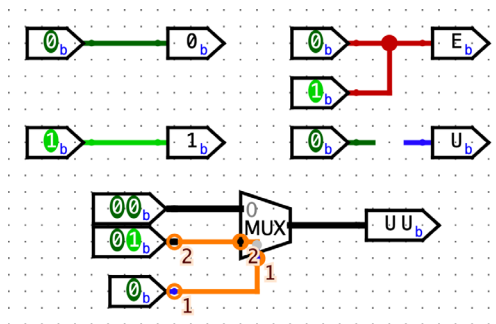


Table 1 Logisim Evolution wire color and meanings.

Color	Meaning
Dark green	A 1-bit wire has a value of 0
Bright green	A 1-bit wire has a value of 1
Black	Multi-bit wire (many components have bit width attributes which can be configured in the attributes menu on the bottom left)

Color	Meaning
Red (values with EEEE) Blue (values with UUUU)	The wire has multiple values on it (in this case, a 0 and 1 from the 2 inputs). Also, remember that a big circle appears at wire junctions. The wire is floating (i.e., has no known value)
Orange	The wire is connected to components that have different bit widths. A simple example is a 1-bit input pin connected to a 2-bit output pin. In the example shown, there's a slightly hidden wire behind the MUX that connects the 2-bit lower data line to the 1-bit select line. Watch out for these!

Combinational Analysis

In the Window menu -> combinational analysis, the Logisim Evolution provides excellent access for combinational logic synthesis from a truth table or logic expression to actual combinational gates.

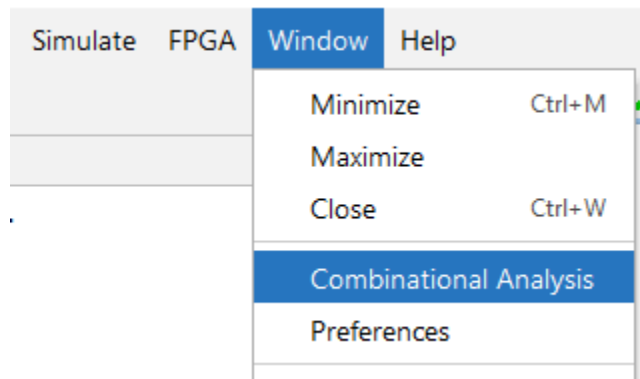


Figure 33 Location of combinational analysis

Once you type in your logic variables, you are allowed to change the truth table, expression, or K-map.

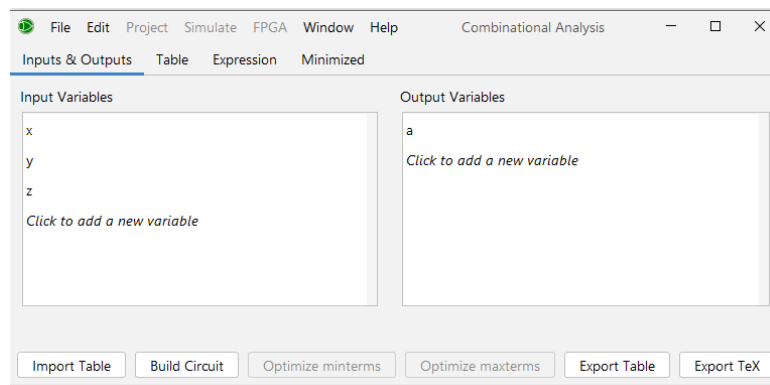
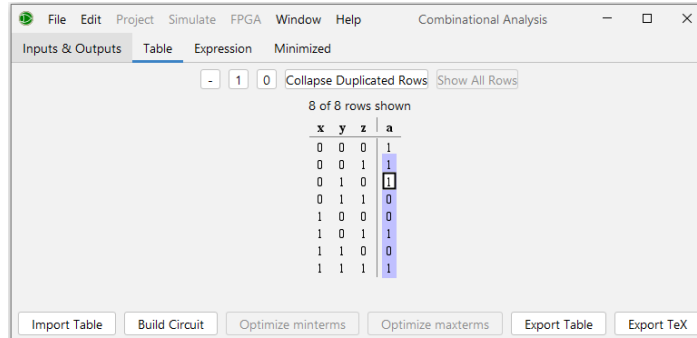


Figure 34 Way to add variables

Logisim Evolution Tutorial



Inputs & Outputs Table Expression Minimized

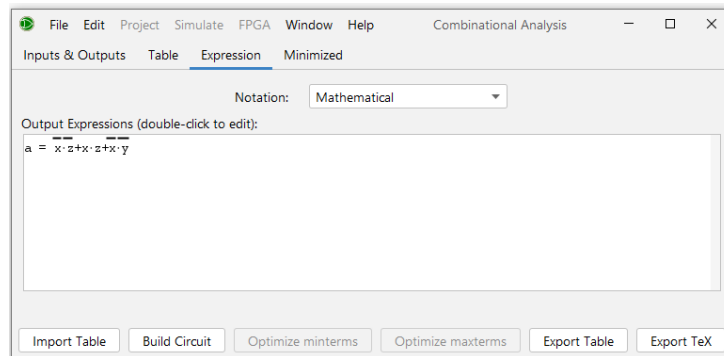
- 1 0 Collapse Duplicated Rows Show All Rows

8 of 8 rows shown

x	y	z	a
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Import Table Build Circuit Optimize minterms Optimize maxterms Export Table Export TeX

Figure 35 Craft desired logic function.



Inputs & Outputs Table Expression Minimized

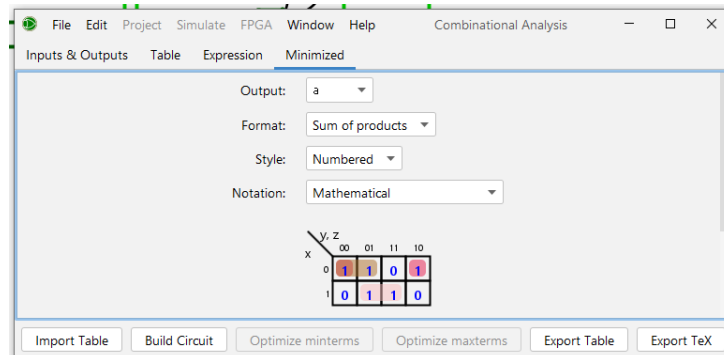
Notation: Mathematical

Output Expressions (double-click to edit):

a = $x'z + xz + xy$

Import Table Build Circuit Optimize minterms Optimize maxterms Export Table Export TeX

Figure 36 Expression result.



Inputs & Outputs Table Expression Minimized

Output: a

Format: Sum of products

Style: Numbered

Notation: Mathematical

		y, z			
		00	01	11	10
x	0	1	1	0	1
	1	0	1	1	0

Import Table Build Circuit Optimize minterms Optimize maxterms Export Table Export TeX

Figure 37 K-map result.

Then you click build circuit, you can then customize how your circuit is built, and name the circuit.

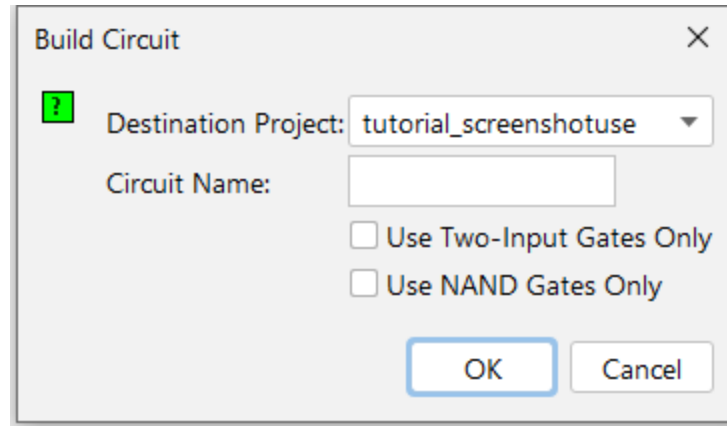
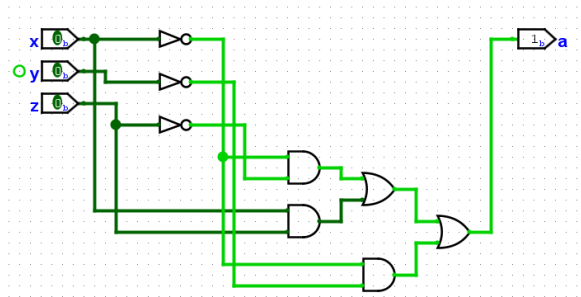


Figure 38 Extract the built digital logic.

The software gives you the tools to do some of the logic designs, while you still need to learn and understand the underlying principles to pass your exam.



Multiple schematics in one .circ file:

For practical use, you may need to save multiple digital logic circuits in a single .circ file to avoid creating multiple .circ files for one lab. Also, you can use the macro for larger designs in circuit design. Such an approach is like what you did in software programming using the concept of modular design by instantiating the functions.

Add a new schematic

You can add new schematic by clicking the + sign here. It allows you to include multiple possible independent circuits within one circuit file, reducing the number of files within your lab or class folders.

Logisim Evolution Tutorial

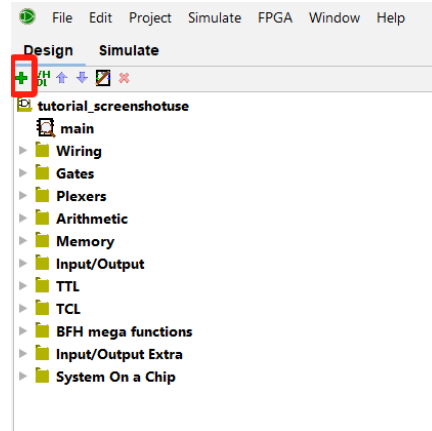


Figure 39 Location of add a new subcircuit.

Then you are asked to name the new circuit, here, let us say and2 for the 2-input AND gate circuit.

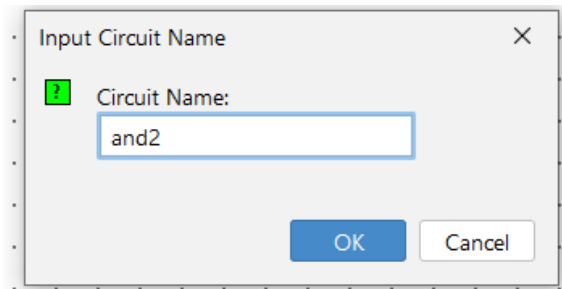


Figure 40 Name the circuit.

Then you will see there is an and2 circuit beneath the default main schematic. It is not necessary to have the top level under the “main” circuit.

Then let's have another 2-input AND gate here. You must use input pins for inputs and output pins for outputs to enable connectivity to and from the subcircuit.

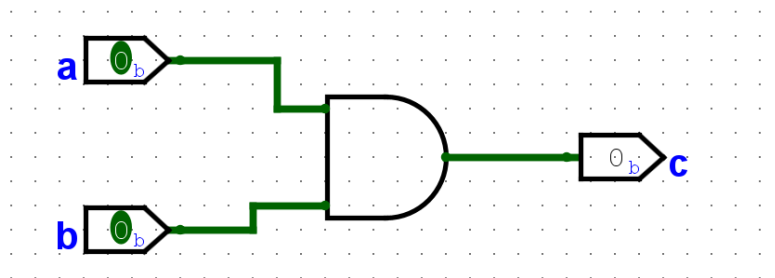
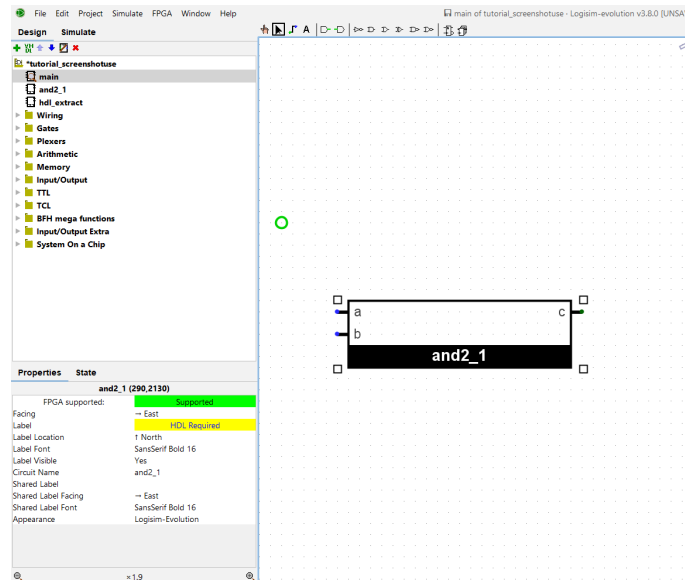


Figure 41 2 input and gate with label on IOs.

Subcircuit/Macro

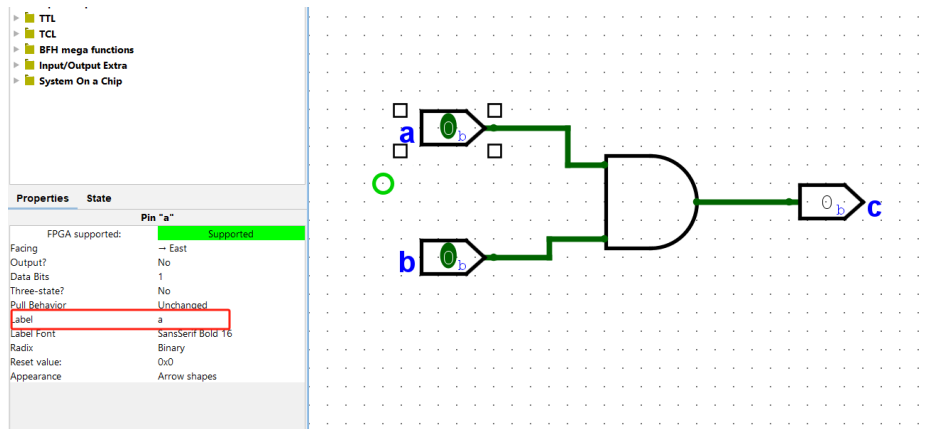
Go back to the main circuit, drag the and2_1 to the main circuit as below.

Logisim Evolution Tutorial

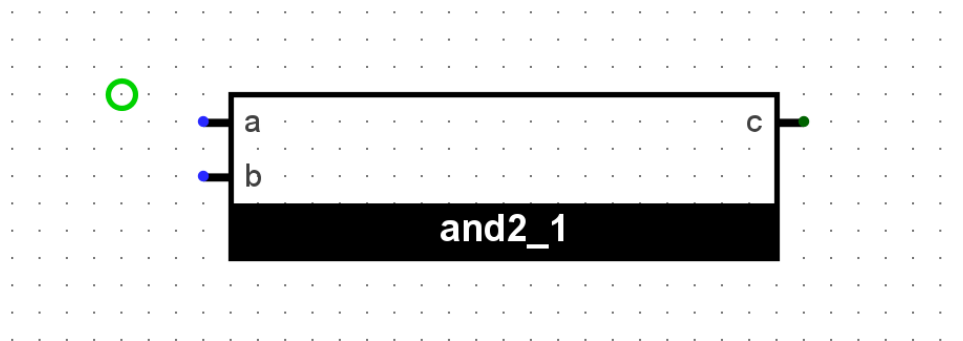


By default, the left part will be input pins, and the right part will be output pins. Please make sure your input and output pins are ordered correctly.

It is highly recommended to edit the input pins and output pins by changing the label property as shown below.



So that there will be clear labels shown on the macro. You can then connect it with the input and output pins to verify its functionality.



You can also further rename and label the subcircuit by yourself.

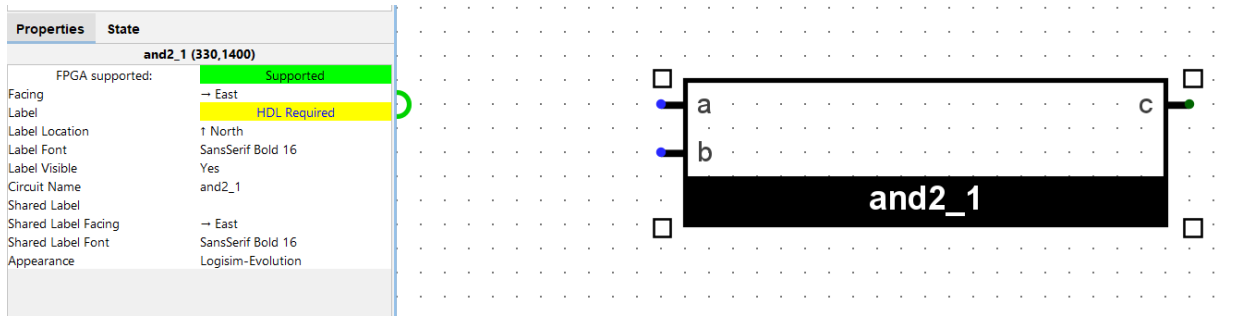


Figure 42 Pack a 2-input AND gate as a macro.

Memory

In this course, students work with memory components, including their use as a cache. In the memory folder, you can see the available devices that can serve as memory.

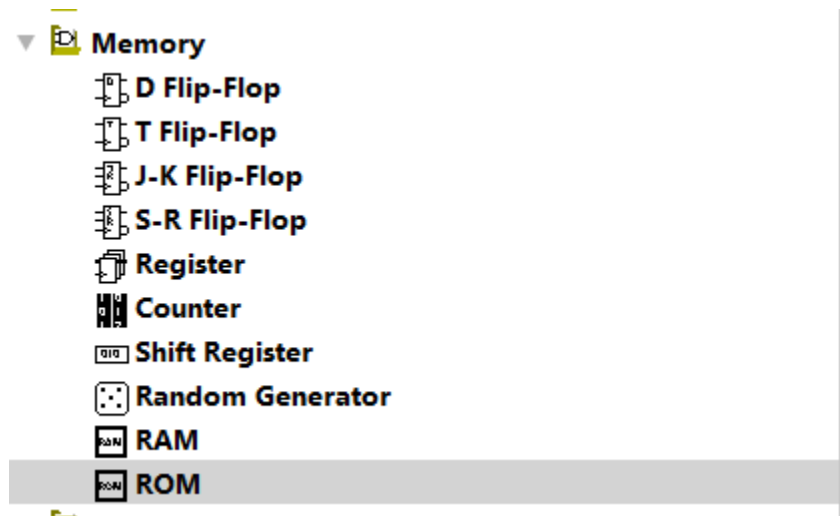


Figure 43 Location of ROM/RAM.

This section covers only the ROM; RAM is similar but includes write enable, output enable, and clock pins. Note that the numbers circled in red are ROM/RAM addresses, not data.

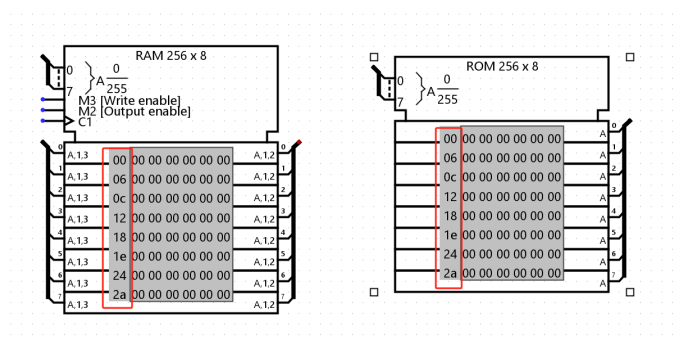


Figure 44 RAM & ROM, addresses are circled in red.

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Once you have placed the ROM/RAM, you can see the stored data shown in the grey table. You can configure the data bits (by default, it is 8 bits) or data storage depth (by default, it is 256) to the value you want by changing the address bit width and data bit width.

Properties	State
ROM (370,2430)	
FPGA supported:	Supported
Address Bit Width	8
Data Bit Width	8
Line size	Single
Allow misaligned?	No
Contents	(click to edit)
Label	HDL Required
Label Font	SansSerif Bold 16
Label Visible	No
Appearance	Logisim-Evolution

Figure 45 Property of ROM.

You can directly edit the content by right-clicking the grey box or clicking the contents in properties. In default, the editing is in HEX.

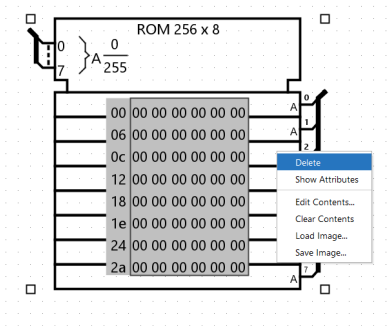


Figure 46 Edit the content of ROM.

As shown in the figure below, I changed the content in address 0 to 0x0f.

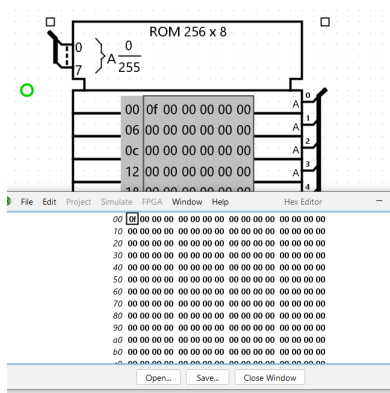


Figure 47 Editing window of ROM.

Sequential logic

Another important category of digital logic is sequential logic. For sequential logic, you need a clock signal to observe its behavior.

Clock signal

You can find the clock signal under the wiring folder.

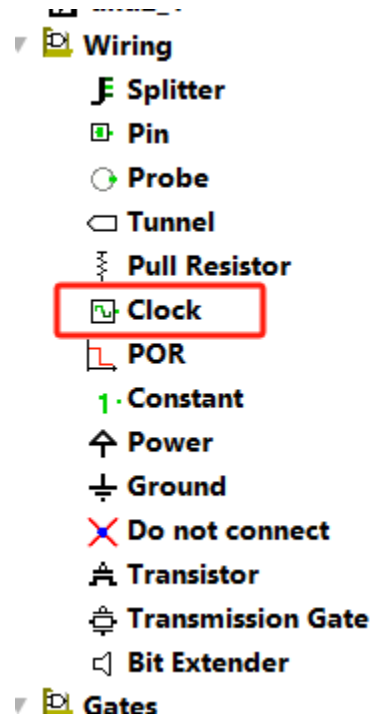


Figure 48 Location of the clock signal.

Then you can connect your clock signal to a sequential logic, here shown is an example for a D flip flop:

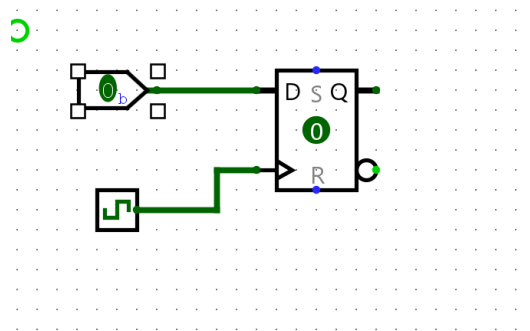


Figure 49 Example connection of clk to a D flip-flop.

You can see the internal state of the D flip-flop, and change it if needed.

Similarly, for sequential logic chips, like 74161, once you make the internal structure visible, you can also observe the internal state and possibly change it, as shown below:

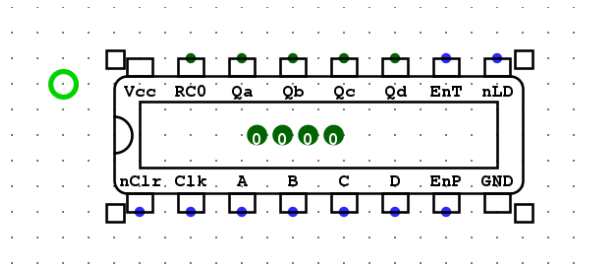


Figure 50 Interior of integrated counter.

You can toggle the clock signal as you toggle combinational pins. However, it may not be efficient for large circuits.

Auto tick clock signal

You can then go to the simulation to set the auto-tick frequency. Typically, around 4 Hz is the best practice. After you set the frequency and enable auto tick, you may see that all clock signals in the current circuit tick according to the frequency.

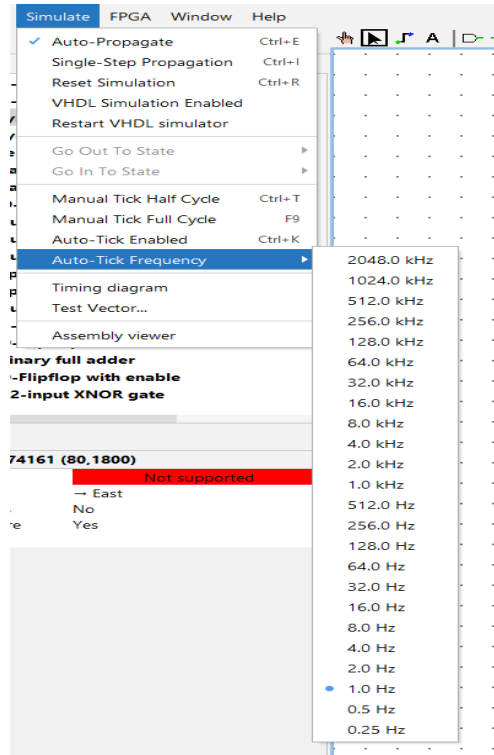


Figure 51 Change the simulation clock frequency.

You can also observe the ticking pattern from the clock signal.



Figure 52 Frequency of operating clock signal.

One issue with the clock signal is that its icon does not change based on orientation. Please always ensure you are actually connected to the clock signal pin.

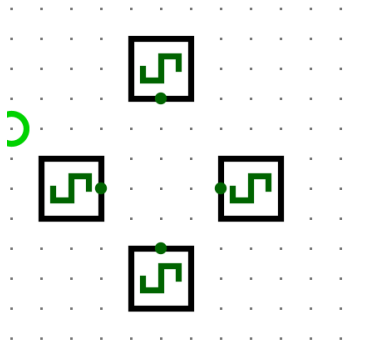


Figure 53 Clock signal with different directions.

Digital Oscilloscope

The digital oscilloscope located under the input/output extra gives you an initial taste of verifying the digital waveform's functionality. Here, the oscilloscope is limited to a binary waveform, which could be extended to decimal/hexadecimal representation in real EDA software.

Connect the oscilloscope to the clock signal, then to the signal you want to observe. You can also easily configure the number of inputs and states displayed by modifying the corresponding properties. It is very useful to observe state transitions in sequential digital logic. Below is a simple connection between a digital oscilloscope and an integrated counter.

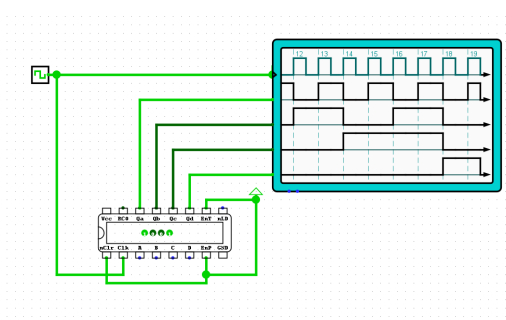
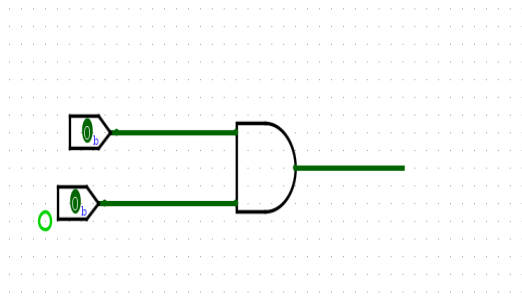


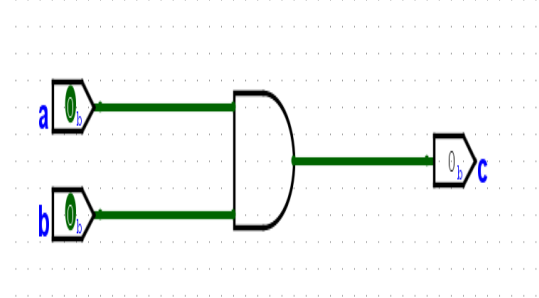
Figure 54 Digital Oscilloscope

Common errors

- No output/input pin provided in the schematic, you need to label them to help you and others know the pin's corresponding functionality, especially when pins are not arranged in a standard or intuitive order."

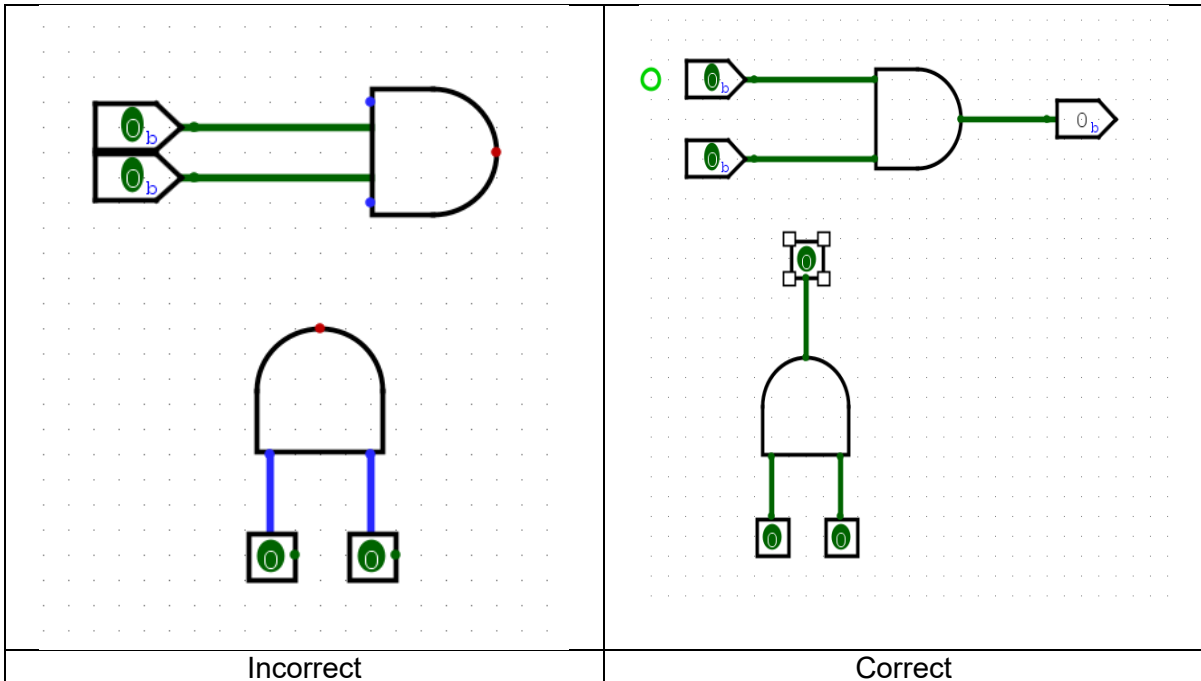


Incorrect

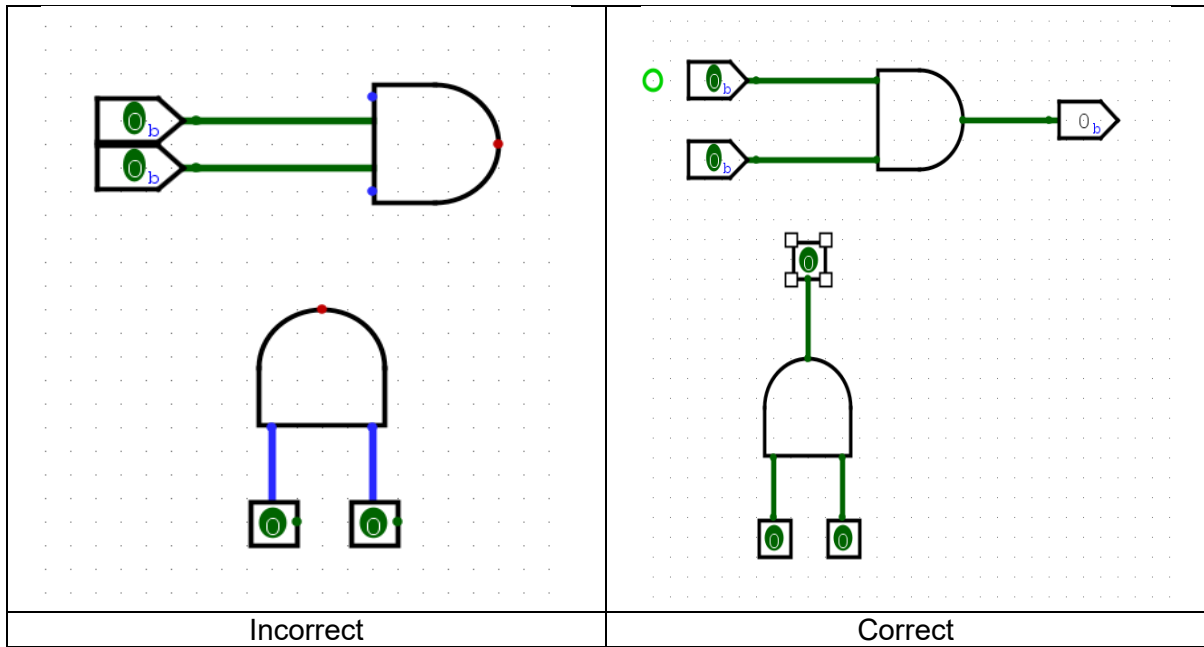


Correct

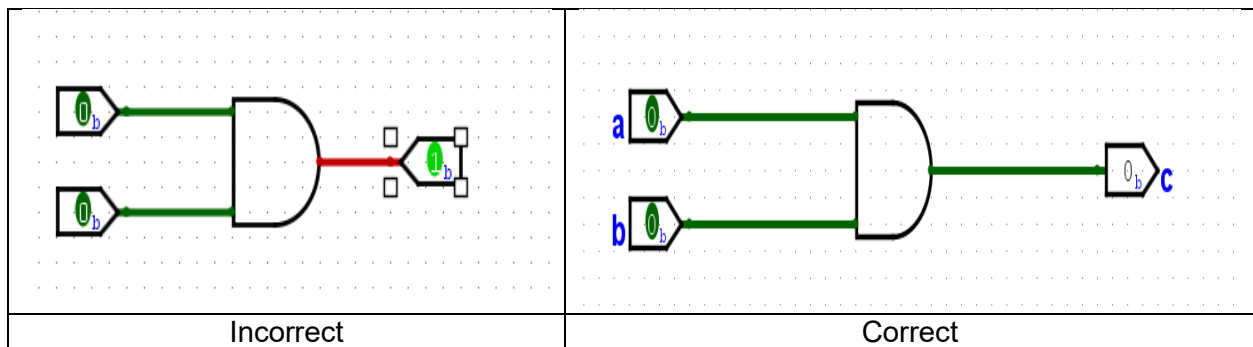
- Wire is not properly connected to the gate/desired pins.



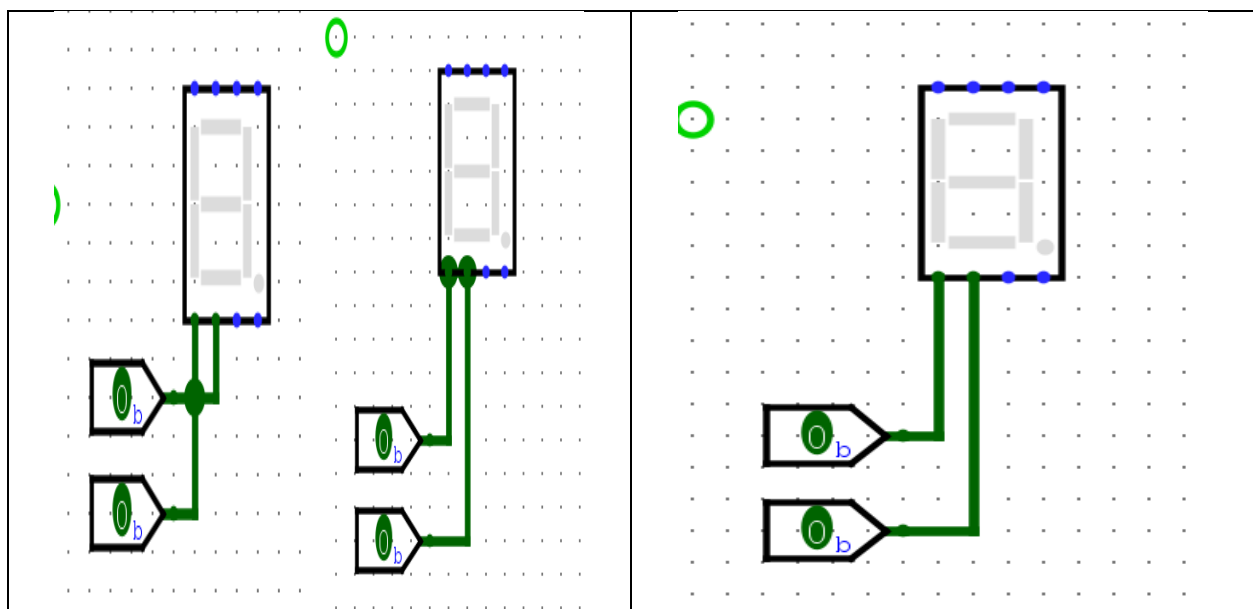
- Wrong circuit in the simulation.

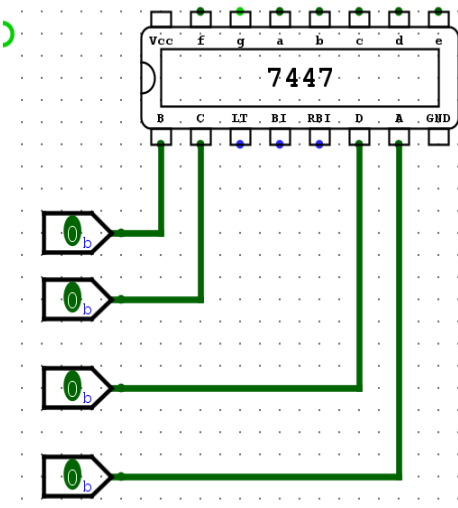
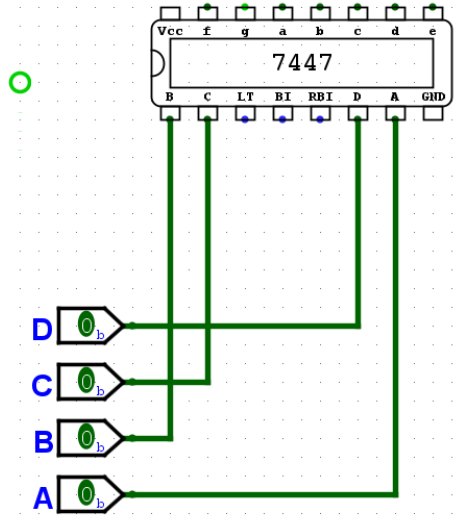


Using an input pin as an output.



Connecting multiple wires together without a driving source.



Incorrect	Correct
Missing necessary labels.	
	
Bad practice	Good practice